4. Performance Evaluation:

- Measure the quality of summaries using evaluation metrics such as:
    - **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** Compares overlaps between generated summaries and reference summaries.
    - **BLEU (Bilingual Evaluation Understudy):** Assesses the quality of generated text by comparing it to human-written text.
- Conduct user surveys to gather qualitative feedback on clarity, accuracy, and relevance.
- Perform A/B testing to identify the most user-preferred summarization models.

5. Deployment:

- Develop a user-friendly web application with:
    - Options to upload text files, paste content, or input URLs for summarization.
    - Real-time feedback and results display.
- Use frameworks like Flask or FastAPI for backend services and React or Vue.js for front-end development.
- Integrate cloud services like AWS or Azure for scalability and faster inference times.
- Include user authentication features for saving summaries or processing confidential documents.

6. Advanced Features:

- Enable language detection and support multilingual summarization.
- Provide summarization options like “brief” or “detailed” based on user preferences.